Linear Cryptanalysis

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June 11, 2026

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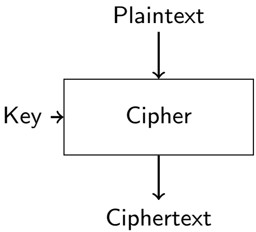
- Cryptographic algorithms are designed to appear random.
- Security depends on the absence of exploitable patterns.
- Cryptanalysis aims to detect and exploit such patterns.
- Successful attacks help evaluate the strength of a cipher.

Goal

Find deviations from random behavior that can reveal information about the secret key.

Linear Cryptanalysis: Basic Idea

- Known-plaintext attack.
- Finds approximate linear relations among:
  - Plaintext bits
  - Ciphertext bits
  - Key bits
- Uses statistical bias to distinguish correct key guesses.



$$P_1 \oplus P_3 \oplus C_2 \oplus K_5 = 0$$

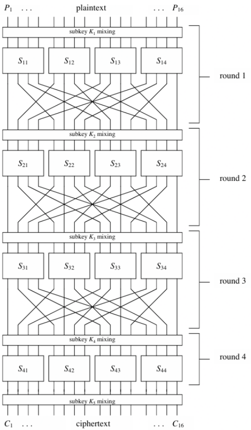
$$Pr(L(P, C, K) = 0) = \frac{1}{2} + \epsilon$$

where ϵ is called the **bias**.

$$Pr = 0.55$$

(Random expectation: 0.50)

Substitution Permutation Network



Linear Approximation and Bias I

Goal: Find a linear relation between input and output bits of an S-box.

$$X = (x_1, x_2, x_3, x_4) \longrightarrow \boxed{\text{S-box}} \longrightarrow Y = (y_1, y_2, y_3, y_4)$$

input	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
output	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

Table 1. S-box Representation (in hexadecimal)

Example approximation:

$$x_2 \oplus x_3 \oplus y_1 \oplus y_3 \oplus y_4 = 0$$

Linear Approximation and Bias II

X_1	X_2	X_3	X_4	Y_1	Y_2	Y_3	Y_4	$X_2 \oplus X_3$	$Y_1 \oplus Y_3 \oplus Y_4$	$X_1 \oplus X_4$	Y_2	$X_3 \oplus X_4$	$Y_1 \oplus Y_4$
0	0	0	0	1	1	1	0	0	0	0	1	0	1
0	0	0	1	0	1	0	0	0	0	1	1	1	0
0	0	1	0	1	1	0	1	1	0	0	1	1	0
0	0	1	1	0	0	0	1	1	1	1	0	0	1
0	1	0	0	0	0	1	0	1	1	0	0	0	0
0	1	0	1	1	1	1	1	1	1	1	1	1	0
0	1	1	0	1	0	1	1	0	1	0	0	1	0
0	1	1	1	1	0	0	0	0	1	1	0	0	1
1	0	0	0	0	0	1	1	0	0	1	0	0	1
1	0	0	1	1	0	1	0	0	0	0	0	1	1
1	0	1	0	0	1	1	0	1	1	1	1	1	0
1	0	1	1	1	1	0	0	1	1	0	1	0	1
1	1	0	0	0	1	0	1	1	1	1	1	0	1
1	1	0	1	1	0	0	1	1	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1	0	1	0
1	1	1	1	0	1	1	1	0	0	0	1	0	1

Table 3. Sample Linear Approximations of S-box

It holds for 12 out of 16 inputs.

$$Pr = \frac{12}{16} = 0.75$$

Bias:

$$\epsilon = Pr - \frac{1}{2} = 0.75 - 0.50 = 0.25$$

Linear Approximation and Bias III

Observation

A non-zero bias indicates a useful linear approximation. Larger $|\epsilon|$ implies a stronger attack.

Question:
How can we systematically find strong approximations?

Answer: The Linear Approximation Table (LAT).

Linear Approximation Table (LAT) I

Definition: The LAT records the bias of all possible linear approximations of an S-box.
For input mask α and output mask β:

$$\alpha \cdot x \oplus \beta \cdot S(x) = 0$$

Meaning of the masks:

$$\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$$

selects input bits and

$$\beta = (\beta_1, \beta_2, \beta_3, \beta_4)$$

selects output bits.
For example,

$$\alpha = 0110 \text{ and } \beta = 1011$$

indicates

$$\alpha \cdot x = x_2 \oplus x_3 \text{ and } \beta \cdot S(x) = y_1 \oplus y_3 \oplus y_4.$$

Linear Approximation Table (LAT) II

Thus the approximation from the previous slide

$$x_2 \oplus x_3 = y_1 \oplus y_3 \oplus y_4$$

corresponds to the mask pair

$$(\alpha, \beta) = (0110, 1011).$$

The LAT entry is

$$LAT[\alpha, \beta] = \#\{x : \alpha \cdot x = \beta \cdot S(x)\} - \frac{N}{2}$$

where $N = 2^n$.

Interpretation

Large positive or negative values indicate strong linear approximations.

Linear Approximation Table (LAT) III

The LAT contains one entry for every possible mask pair (α, β). The largest absolute entries correspond to the strongest linear approximations.

The LAT for the considered 4-bit S-box is shown below.

Input Sum \ Output Sum	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	+8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	−2	−2	0	0	−2	+6	+2	+2	0	0	+2	+2	0	0
2	0	0	−2	−2	0	0	−2	−2	0	0	+2	+2	0	0	−6	+2
3	0	0	0	0	0	0	0	0	+2	−6	−2	−2	+2	+2	−2	−2
4	0	+2	0	−2	−2	−4	−2	0	0	−2	0	+2	+2	−4	+2	0
5	0	−2	−2	0	−2	0	+4	+2	−2	0	−4	+2	0	−2	−2	0
6	0	+2	−2	+4	+2	0	0	+2	0	−2	+2	+4	−2	0	0	−2
7	0	−2	0	+2	+2	−4	+2	0	−2	0	+2	0	+4	+2	0	+2
8	0	0	0	0	0	0	0	0	−2	+2	+2	−2	+2	−2	−2	−6
9	0	0	−2	−2	0	0	−2	−2	−4	0	−2	+2	0	+4	+2	−2
A	0	+4	−2	+2	−4	0	+2	−2	+2	+2	0	0	+2	+2	0	0
B	0	+4	0	−4	+4	0	+4	0	0	0	0	0	0	0	0	0
C	0	−2	+4	−2	−2	0	+2	0	+2	0	+2	+4	0	+2	0	−2
D	0	+2	+2	0	−2	+4	0	+2	−4	−2	+2	0	+2	0	0	+2
E	0	+2	+2	0	−2	−4	0	+2	−2	0	0	−2	−4	+2	−2	0
F	0	−2	−4	−2	−2	0	+2	0	0	−2	+4	−2	−2	0	+2	0

Properties of the Linear Approximation Table (LAT)

Let $S : \mathbb{F}_2^n \to \mathbb{F}_2^n$ be an n-bit bijective S-box.

Important Properties of the LAT

1. **Zero-mask entry**
$$LAT(0,0) = 2^{n-1}.$$

2. **First row and first column**
$$LAT(0,\beta) = 0, \qquad \forall\, \beta \neq 0,$$
$$LAT(\alpha,0) = 0, \qquad \forall\, \alpha \neq 0.$$

3. **Even-valued entries**
$$LAT(\alpha,\beta) \in 2\mathbb{Z}.$$

Implication: Large absolute LAT values correspond to strong linear approximations and are desirable for linear cryptanalysis.

Selecting the Best Linear Approximation

- Compute the LAT.
- Find entries with the largest absolute values.
- These correspond to the strongest biases.
- Use them to construct multi-round approximations.

$$LAT(0110, 1011) = 4$$

$$\epsilon = \frac{4}{16} = 0.25$$

Input Sum \ Output Sum	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	+8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	−2	−2	0	0	−2	+6	+2	+2	0	0	+2	+2	0	0
2	0	0	−2	−2	0	0	−2	−2	0	0	+2	+2	0	0	−6	+2
3	0	0	0	0	0	0	0	0	+2	−6	−2	−2	+2	+2	−2	−2
4	0	+2	0	−2	−2	−4	−2	0	0	−2	0	+2	+2	−4	+2	0
5	0	−2	−2	0	−2	0	+4	+2	−2	0	−4	+2	0	−2	−2	0
6	0	+2	−2	+4	+2	0	0	+2	0	−2	+2	+4	−2	0	0	−2
7	0	−2	0	+2	+2	−4	+2	0	−2	0	+2	0	+4	+2	0	+2
8	0	0	0	0	0	0	0	0	−2	+2	+2	−2	+2	−2	−2	−6
9	0	0	−2	−2	0	0	−2	−2	−4	0	−2	+2	0	+4	+2	−2
A	0	+4	−2	+2	−4	0	+2	−2	+2	+2	0	0	+2	+2	0	0
B	0	+4	0	−4	+4	0	+4	0	0	0	0	0	0	0	0	0
C	0	−2	+4	−2	−2	0	+2	0	+2	0	+2	+4	0	+2	0	−2
D	0	+2	+2	0	−2	+4	0	+2	−4	−2	+2	0	+2	0	0	+2
E	0	+2	+2	0	−2	−4	0	+2	−2	0	0	−2	−4	+2	−2	0
F	0	−2	−4	−2	−2	0	+2	0	0	−2	+4	−2	−2	0	+2	0

Piling-Up Lemma

Suppose r linear approximations have biases

$$\epsilon_1, \epsilon_2, \ldots, \epsilon_r.$$

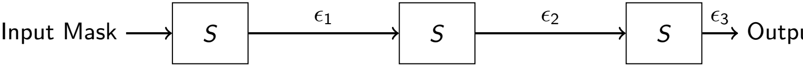
Then the combined approximation has bias

$$\epsilon = 2^{r-1} \prod_{i=1}^{r} \epsilon_i.$$

Observation

Bias decreases rapidly as the number of rounds increases. Therefore, strong single-round approximations are crucial.

Example: A 3-Round Linear Characteristic



Suppose

$$\epsilon_1 = \frac{1}{4}, \qquad \epsilon_2 = \frac{1}{8}, \qquad \epsilon_3 = \frac{1}{4}.$$

Using the Piling-Up Lemma,

$$\epsilon = 2^2 \left(\frac{1}{4}\right) \left(\frac{1}{8}\right) \left(\frac{1}{4}\right) = \frac{1}{32}.$$

Therefore, the resulting approximation holds with probability

$$Pr = \frac{1}{2} + \frac{1}{32}.$$

From S-box Bias to Key Recovery

Linear Approximation Table (LAT)

↓

Best S-box Approximation

↓

3-Round Linear Characteristic

↓

Approximation with Non-zero Bias

↓

Guess Last-Round Key Bits

↓

Partial Decryption

↓

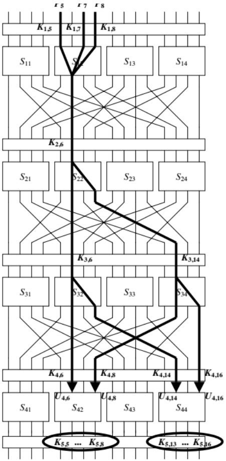
Bias Counting

↓

Recover Subkey

Goal

The 4-Round SPN Cipher I



S_{12}: $X_1 \oplus X_3 \oplus X_4 = Y_2$ with probability 12/1
S_{22}: $X_2 = Y_2 \oplus Y_4$ with probability 4/16
S_{32}: $X_2 = Y_2 \oplus Y_4$ with probability 4/16
S_{34}: $X_2 = Y_2 \oplus Y_4$ with probability 4/16

U_i: Input to the round i s-box and V_i: output of round i s-box

Selecting a Strong S-box Approximation

From the LAT we choose

$$X_1 \oplus X_3 \oplus X_4 = Y_2$$

It holds for

12 out of 16

inputs.

$$Pr = \frac{12}{16} = 0.75$$

Bias:

$$\epsilon = 0.75 - 0.50 = \frac{1}{4}$$

Observation

Large bias implies a useful approximation for cryptanalysis.

We know $U_1 = P \oplus K_1$ and the linear approximation in the first round is

$$\begin{aligned} V_{1,6} &= U_{1,5} \oplus U_{1,7} \oplus U_{1,8} \\ &= P_5 \oplus K_{1,5} \oplus P_7 \oplus K_{1,7} \oplus P_8 \oplus K_{1,8} \end{aligned}$$

occurs with probability $\frac{3}{4}$.

We know $U_1 = P \oplus K_1$ and the linear approximation in the first round is

$$\begin{aligned} V_{1,6} &= U_{1,5} \oplus U_{1,7} \oplus U_{1,8} \\ &= P_5 \oplus K_{1,5} \oplus P_7 \oplus K_{1,7} \oplus P_8 \oplus K_{1,8} \end{aligned}$$

occurs with probability $\frac{3}{4}$.
Approximation in the second round

$$U_{2,6} = V_{2,6} \oplus V_{2,8}$$
$$V_{1,6} \oplus K_{2,6} = V_{2,6} \oplus V_{2,8}$$

We know $U_1 = P \oplus K_1$ and the linear approximation in the first round is

$$\begin{aligned} V_{1,6} &= U_{1,5} \oplus U_{1,7} \oplus U_{1,8} \\ &= P_5 \oplus K_{1,5} \oplus P_7 \oplus K_{1,7} \oplus P_8 \oplus K_{1,8} \end{aligned}$$

occurs with probability $\frac{3}{4}$.
Approximation in the second round

$$\begin{aligned} U_{2,6} &= V_{2,6} \oplus V_{2,8} \\ V_{1,6} \oplus K_{2,6} &= V_{2,6} \oplus V_{2,8} \end{aligned}$$

holds with probability $\frac{1}{4}$. Combining both the rounds

$$V_{2,6} \oplus V_{2,8} \oplus P_5 \oplus P_7 \oplus P_8 \oplus K_{1,5} \oplus K_{1,7} \oplus K_{1,8} \oplus K_{2,6} = 0 \qquad (1)$$

with probability $\frac{3}{8}$.

3rd round Approximation

Approximation of round 3

$$V_{3,6} \oplus V_{3,8} = U_{3,6} \text{ holds with probability } \frac{1}{4}$$

$$V_{3,14} \oplus V_{3,16} = U_{3,14} \text{ holds with probability } \frac{1}{4}$$

3rd round Approximation

Approximation of round 3

$$V_{3,6} \oplus V_{3,8} = U_{3,6} \text{ holds with probability } \frac{1}{4}$$

$$V_{3,14} \oplus V_{3,16} = U_{3,14} \text{ holds with probability } \frac{1}{4}$$

We know $U_{3,6} = V_{2,6} \oplus K_{3,6}$ and $U_{3,14} = V_{2,8} \oplus K_{3,14}$.

3rd round Approximation

Approximation of round 3

$$V_{3,6} \oplus V_{3,8} = U_{3,6} \text{ holds with probability } \frac{1}{4}$$

$$V_{3,14} \oplus V_{3,16} = U_{3,14} \text{ holds with probability } \frac{1}{4}$$

We know $U_{3,6} = V_{2,6} \oplus K_{3,6}$ and $U_{3,14} = V_{2,8} \oplus K_{3,14}$. Therefore,

$$V_{3,6} \oplus V_{3,8} \oplus V_{3,14} \oplus V_{3,16} \oplus V_{2,6} \oplus K_{3,6} \oplus V_{2,8} \oplus K_{3,14} = 0 \qquad (2)$$

with probability $\frac{1}{8}$.

3 round Approximation

Combining the approimation of three rounds

$$V_{3,6} \oplus V_{3,8} \oplus V_{3,14} \oplus V_{3,16} \oplus K_{3,6} \oplus K_{3,14} \oplus P_5 \oplus P_7 \oplus P_8 \oplus K_{1,5} \oplus K_{1,7} \oplus K_{1,8} \oplus K_{2,6} = 0$$

3 round Approximation

Combining the approimation of three rounds

$$V_{3,6} \oplus V_{3,8} \oplus V_{3,14} \oplus V_{3,16} \oplus K_{3,6} \oplus K_{3,14} \oplus P_5 \oplus P_7 \oplus P_8 \oplus K_{1,5} \oplus K_{1,7} \oplus K_{1,8} \oplus K_{2,6} = 0$$

We know $U_{4,6} = V_{3,6} \oplus K_{4,6}$, $U_{4,8} = V_{3,14} \oplus K_{4,8}$, $U_{4,14} = V_{3,8} \oplus K_{4,14}$ and $U_{4,16} = V_{3,16} \oplus K_{4,16}$

3 round Approximation

Combining the approimation of three rounds

$V_{3,6} \oplus V_{3,8} \oplus V_{3,14} \oplus V_{3,16} \oplus K_{3,6} \oplus K_{3,14} \oplus P_5 \oplus P_7 \oplus P_8 \oplus K_{1,5} \oplus K_{1,7} \oplus K_{1,8} \oplus K_{2,6} = 0$

We know $U_{4,6} = V_{3,6} \oplus K_{4,6}$, $U_{4,8} = V_{3,14} \oplus K_{4,8}$, $U_{4,14} = V_{3,8} \oplus K_{4,14}$ and $U_{4,16} = V_{3,16} \oplus K_{4,16}$ Therefore,

$$U_{4,6} \oplus U_{4,8} \oplus U_{4,14} \oplus U_{4,16} \oplus \Sigma_K = 0$$

where

$\Sigma_K = K_{4,6} \oplus K_{4,8} \oplus K_{4,14} \oplus K_{4,16} \oplus K_{1,5} \oplus K_{1,7} \oplus K_{1,8} \oplus K_{2,6} \oplus K_{3,6} \oplus K_{3,14}$

holds with probability $\frac{15}{32}$.

Overall Bias

Four active approximations:

$$\epsilon_1 = \frac{1}{4}$$

$$\epsilon_2 = \epsilon_3 = \epsilon_4 = -\frac{1}{4}$$

Using the Piling-Up Lemma,

$$\epsilon = 2^3 \left(\frac{1}{4}\right) \left(-\frac{1}{4}\right)^3 = -\frac{1}{32}.$$

Hence

$$Pr = \frac{1}{2} - \frac{1}{32} = \frac{15}{32}.$$

Why Partial Decryption?

Known:

$$(P, C)$$

Unknown:

$$U_{4,6}, U_{4,8}, U_{4,14}, U_{4,16}.$$

These bits occur before the last round.

Idea

Guess a small number of last-round key bits and partially decrypt the ciphertext.

Why Partial Decryption?

Known:

$$(P, C)$$

Unknown:

$$U_{4,6}, U_{4,8}, U_{4,14}, U_{4,16}.$$

These bits occur before the last round.

Idea

Guess a small number of last-round key bits and partially decrypt the cipher-text.

The approximation involves

$$S_{42} \quad \text{and} \quad S_{44}.$$

Therefore we guess $[K_{5,5} \ldots K_{5,8}, K_{5,13} \ldots K_{5,16}]$
Total key guesses:$2^8 = 256$.

Key Recovery Algorithm

For each key guess:

1. Guess 8 key bits.
2. Partially decrypt ciphertext.
3. Recover

$$U_{4,6}, U_{4,8}, U_{4,14}, U_{4,16}.$$

4. Evaluate the linear approximation.
5. Update a counter.

Repeat for all plaintext-ciphertext pairs.

Selecting the Correct Key

For a key guess k,

$$T_k = \#\{(P, C) : L(P, U_k) = 0\}.$$

Correct key:

$$T_k \approx N\left(\frac{1}{2} \pm \frac{1}{32}\right).$$

Wrong key:

$$T_k \approx \frac{N}{2}.$$

Decision Rule

Choose the key guess with the largest

$$\left| T_k - \frac{N}{2} \right|.$$

Example: Key Ranking

Suppose $N = 10000$ pairs.

Key Guess	Count T
00	5010
01	4975
10	5512
11	5031

Biases:

$$|5010 - 5000| = 10$$

$$|4975 - 5000| = 25$$

$$|5512 - 5000| = 512$$

$$|5031 - 5000| = 31$$

The key guess 10 is the most likely candidate.

Key Recovery Principle

Goal: Use a linear approximation to identify the correct key bits.

Guess Key Bits

$\Downarrow$

Evaluate the Linear Approximation

$\Downarrow$

Count the Number of Times it Holds

$\Downarrow$

Measure the Bias

$\Downarrow$

Largest Bias $\Rightarrow$ Correct Key Guess

Data Complexity

The number of required plaintext-ciphertext pairs is

$$N \approx \frac{1}{\epsilon^2}.$$

Examples:

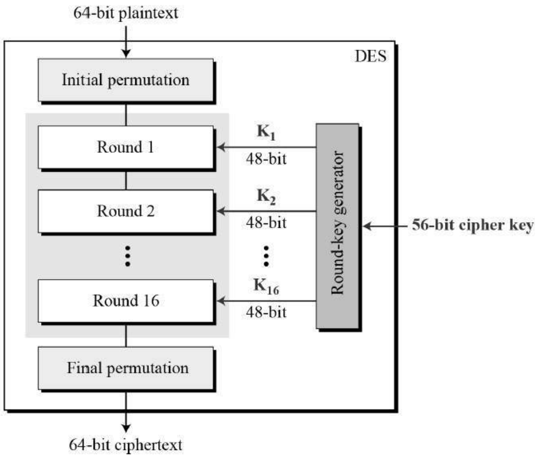
Bias ϵ	Required Data
0.1	100
0.01	10^4
0.001	10^6

Observation

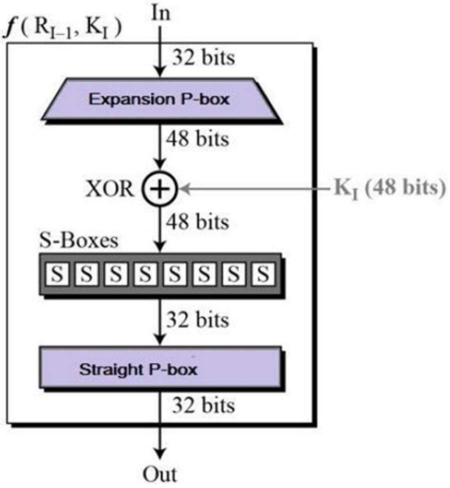
Stronger approximations require fewer data samples.

Linear Cryptanalysis on Feistel Network

Data Encryption Standard



Data Encryption Standard-Round Function



Strategy

How do we find linear approximations in DES?

Strategy

How do we find linear approximations in DES?

We study:

1. A DES S-box.
2. Propagation through the DES f-function.
3. Construction of a 3-round approximation.

DES S-box S_5

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

Observation from the truth table of S_5:

$$x_2 = y_1 \oplus y_2 \oplus y_3 \oplus y_4$$

with probability

$$\frac{52}{64} = 0.8125.$$

DES S-box S_5

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

Observation from the truth table of S_5:

$$x_2 = y_1 \oplus y_2 \oplus y_3 \oplus y_4$$

with probability

$$\frac{52}{64} = 0.8125.$$

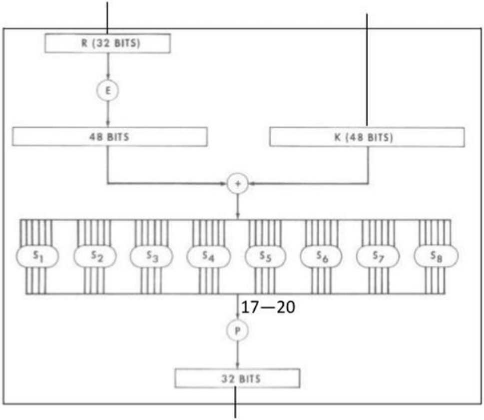
Bias:

$$\varepsilon = \frac{52}{64} - \frac{1}{2} = \frac{20}{64}.$$

Key Point

The S-box is not perfectly random and exhibits exploitable linear behavior.

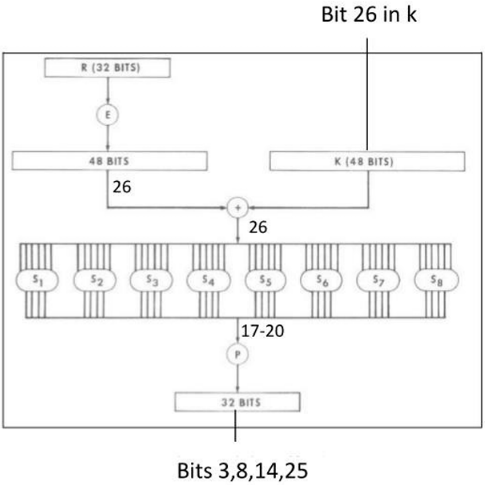
Feistel Function



Permutation P

16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

Feistel Function

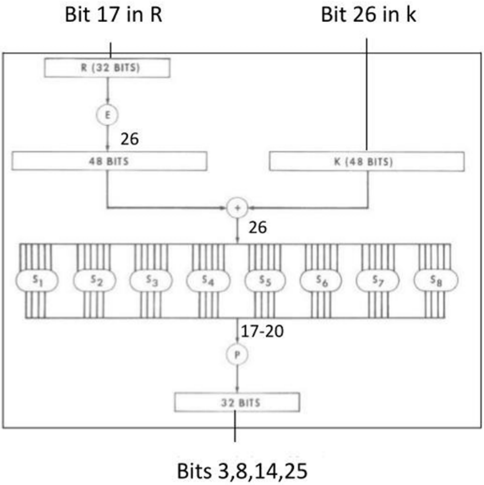


Expansion Function

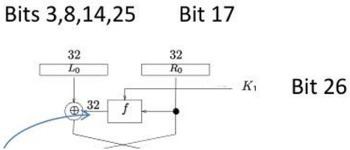
E BIT-SELECTION TABLE

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

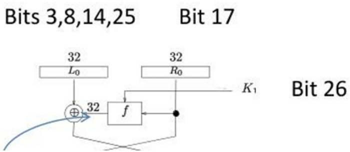
Feistel Function



One Round Approximation



One Round Approximation

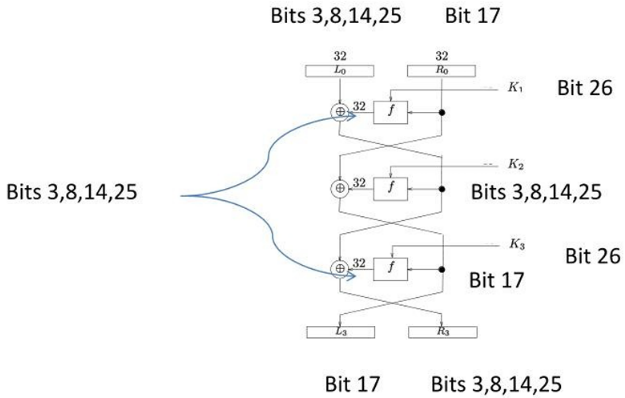


For the first round, the linear approximation is

$$L_0[3] \oplus R_1[3] \oplus L_0[8] \oplus R_1[8] \oplus L_0[14] \oplus R_1[14] \oplus L_0[25] \oplus R_1[25] = K_1[26] \oplus R_0[17]$$

Extension to 3 rounds

Assume the approximation is extended to three rounds



3 round linear approximation

First round linear approximation occurs with probability $\frac{52}{64}$

$$L_0[3]\oplus R_1[3]\oplus L_0[8]\oplus R_1[8]\oplus L_0[14]\oplus R_1[14]\oplus L_0[25]\oplus R_1[25] = K_1[26]\oplus R_0[17]$$

Third round linear approximation occurs with probability $\frac{52}{64}$

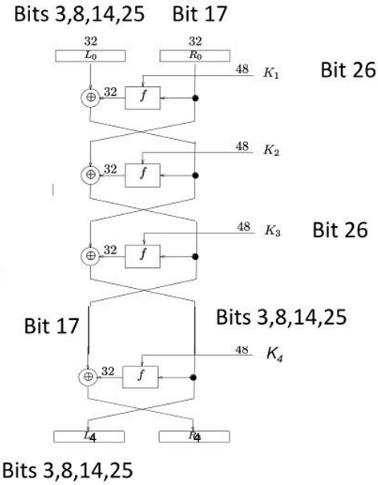
$$R_3[3]\oplus R_1[3]\oplus R_3[8]\oplus R_1[8]\oplus R_3[14]\oplus R_1[14]\oplus R_3[25]\oplus R_1[25] = K_3[26]\oplus L_3[17]$$

Thus,

$$L_0[3] \oplus L_0[8] \oplus L_0[14] \oplus L_0[25] \oplus R_3[3] \oplus R_3[8]\oplus R_3[14] \oplus R_3[25] \oplus R_0[17] \oplus L_3[17] = K_1[26] \oplus K_3[26]$$

holds with probability 0.7 ($\epsilon_1 = \epsilon_2 = \frac{20}{64}$, bias=0.195)

Attack on 4 round DES



4 Round Attack

The approximation is

$$L_0[3] \oplus L_0[8] \oplus L_0[14] \oplus L_0[25] \oplus L_4[3] \oplus L_4[8] \oplus L_4[14] \oplus L_4[25] \oplus R_0[17] \oplus L_3[17] = K_1[26] \oplus K_3[26]$$

We know $L_3[17] = R_4[17] \oplus F(L_4, K_4)[17]$.

- Only 6 subkey bits affect the targeted S-box.
- Try all 2^6 possibilities.
- The correct guess maximizes the observed bias.

Conclusion

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Conclusion

- Linear cryptanalysis is a known-plaintext attack.
- The Linear Approximation Table (LAT) helps identify strong linear approximations of S-boxes.
- The Piling-Up Lemma allows single-round approximations to be combined into multi-round characteristics.
- In SPN ciphers, linear approximations can be used together with partial decryption to recover subkey bits.
- The same principles extend to Feistel ciphers such as DES.
- By guessing a small number of last-round key bits and exploiting the observed bias, key information can be recovered efficiently.

Future Directions

The ideas developed in classical linear cryptanalysis have led to several advanced techniques:

- Multiple Linear Cryptanalysis
- Multidimensional Linear Cryptanalysis
- Differential-Linear Cryptanalysis
- Zero-Correlation Linear Cryptanalysis

Thank You